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1 November 1974

MILITARY SPECIFICATION

TANK, MASTER, X-RAY FILM PROCESSING

FOR 70 MM FILM

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a self-contained, temperature-controlled developing tank for developing 70 mm X-ray film.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
W-C-596	- Connector, Electrical, Power, General Specification For.
BB-F-1421	- Fluorocarbon Refrigerants.
CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.
QQ-S-766	- Steel Plates, Sheets, and Strip-Corrosion Resisting.

FSC 6635

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-621	- Boxes, Wood, Nailed, and Lock-Corner.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-2212	- Controllers, Electric Motors, AC or DC, and Associated Switching Devices, Naval Shipboard.
MIL-R-14328	- Rubber Sheet: Synthetic, Medium Soft, General-Purpose Gasket Material (for Extreme Climatic Conditions).
MIL-P-15024/6	- Plates, Identification, Equipment.

STANDARDS

FEDERAL

FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM A 380 - Cleaning and Descaling Stainless Steel Surfaces.

(Application for copies should be addressed to the American National Standards Institute, Inc. 1430 Broadway, New York, NY 10018).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 North West Seventh Street, Miami, FL 33125.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The developing tank shall consist of a main tank containing circulating water and two insert tanks for the solutions required for developing 70 mm X-ray film. The tank shall contain a refrigeration unit and a heating unit to control the temperature of the circulating water. The complete unit shall be contained in a single cabinet.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

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3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.4 Safety. All moving, reciprocating, or rotating parts, such as but not limited to, fans, motors, or belts shall be enclosed or adequately guarded. All electrical components shall be effectively grounded. Electrical equipment, wires, and connectors shall be effectively guarded or insulated to protect all persons and objects from electrical shock hazards. Excessive length of fastening devices exposed during use or maintenance shall be avoided.

3.5 Environmental requirements.

3.5.1 Operation. The tank shall operate as specified herein in any ambient temperature from plus 40° F to plus 120° F. The refrigeration system or the heating system shall cool or heat, as applicable, the water in the tank from either extreme to a temperature of 68° F, plus or minus 2° F, in a period of not more than 2 hours and shall maintain that temperature automatically.

3.5.2 Storage. The tank shall not be damaged by storage under the following conditions:

Air temperature
High - Plus 160° F for 4 hours daily.
Low - Minus 30° F for a period of 12 hours.
No solar radiation.

3.6 Electrical requirements. The tank shall operate on 115-volt, 60-hertz, single-phase alternating current. Wiring shall be 2-pole, 3-wire grounded system. Maximum power consumption at any one time shall be not more than 1200 watts.

3.6.1 Cord. A flexible electric power cord, not less than 6 feet long, shall be wired to the tank to provide power to the various components. The cord shall terminate with a plug conforming to W-C-596, Series D, Style D21.

3.7 Motors. Motors shall conform to CC-M-1807, Type II, Class 3, Style E, Design N. Motors shall be equipped with permanently lubricated ball bearings.

3.8 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.8.1), with 1 percent copper 8-quinolinolate (by weight) based on the non-volatile content of the varnish.

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(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.8.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish preclude the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.9 Cabinet. The complete film processing tank shall be contained in a cabinet with the following overall measurements:

Width (left to right)	24.5 inches.
Depth (front to back)	26.5 inches.
Working height	39.5 inches.
Overall height (cover closed)	41.5 inches.

A tolerance of plus or minus 0.25 inch will be allowed for all overall measurements. The cabinet shall be supported by four hot-rolled steel angles not less than 1.5 by 0.25 inch. Each angle shall have a flat foot welded to the bottom. The foot shall have a 0.5-inch hole for bolting the cabinet to the floor. The outer cover for the cabinet shall be aluminum conforming to QQ-A-250/11, Temper T4. The top of the cabinet shall support the main water tank. The top of the main water tank and the working surface of the cabinet shall be flush to provide a watertight seal when the cover is closed. The back of the cabinet shall have a vertical mounting flange with three equally spaced 0.375-inch holes for bolting the cabinet to a wall. The lower portion of the cabinet shall contain the refrigeration system, the heating unit, the circulating pump, and the filter. The lower portion of the cabinet shall be equipped with removable panels, where necessary, to permit servicing and replacement of components. Panels shall be secured with quick-disconnect fasteners.

3.10 Main water tank. The inside dimensions of the main water tank shall be as follows:

Width (left to right)	15.5 inches.
Depth (front to back)	18.5 inches.
Height	23.5 inches.

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All inside dimensions of the water tank shall have a tolerance of plus or minus 0.125 inch. The main tank shall consist of an inner liner and an outer case. The top shall be open. The space between the inner liner and the outer case including the bottom shall not be less than 1 inch. The liner and the outer case shall be corrosion-resistant steel conforming to QQ-S-766, Class 316 or 316L. The liner shall be 18 gage and the case shall be 20 gage. The inside of the liner shall have a 2B finish. The space between the liner and outer case shall be insulated with a 2.5-pound high-density rigid polyurethene foam formed in place. A single valve shall be provided to drain, by gravity, all the water in the entire system. The valve shall be placed in the front of the cabinet in an easily accessible position. An overflow pipe 1 inch in diameter shall be placed 2 inches below the top of the water tank, and shall be positioned so that it does not interfere with the insert tanks. The overflow pipe shall be connected to the drainpipe below the drain valve. All seams in the inner liner, the outer case, and pipes entering or leaving the tank shall be welded and shall not leak.

3.10.1 Main tank cover. The main tank cover shall be hinged at the rear, and shall cover the top of the main tank when in the down position. Hinged screw and wing clamps shall be provided to positively lock the cover in place. The cover shall be provided with a gasket conforming to MIL-R-14328, type and class optional, bonded to the cover. When in the closed position and clamped in place, the cover shall provide a lighttight and watertight seal between the cover and the main tank. In addition, the main cover shall hold the insert tanks and their covers in place without leaking when tested as specified in 4.5.2.2. A handle shall be provided to lift the cover to the vertical position. The cover shall be reinforced on each side with 1-inch by 1-inch by 0.189-inch angles which extend to the rear of the tank and serve as hinge points for the cover. The front and rear of the cover shall be flanged upward to a height flush with the reinforcement angles. The hinge point holes shall be slotted out to allow the cover to be pivoted to a vertical position or to completely remove the cover from the tank. When in the vertical position, the cover shall rest against the wall behind the tank (when the tank is mounted in the laboratory), and shall remain open by its own weight when the laboratory is level. Cam-type bolts shall be provided on the uprights to permit final adjustment and to compensate for gasket wear. The entire cover, reinforcing angles, uprights, and cam-type bolts shall be of corrosion-resisting steel conforming to QQ-S-766, Class 316 or 316L, Finish 2B. The cover shall be 12-gage steel.

3.10.2 Cover latch. A separate latch shall be provided with each tank to hold the cover in a vertical open position against the back wall when the laboratory is in a tilted position. The latch shall mount on the wall after the tank has been installed. The latch shall automatically engage the handle of the cover when the cover is pushed against it. A slight touch shall release the cover. The latch shall be the same material as the cover, and shall be provided with wood screws to secure it to a 0.625-inch plywood wall.

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3.11 Insert tanks. Two insert tanks shall be furnished with each main tank. The insert tanks shall be 16-gage corrosion-resisting steel conforming to QQ-S-766, Class 316 or 316L, Finish 2B. Each insert tank shall be rectangular with an open top. The top edge shall be flanged outward at right angles to the sides and ends 0.25 inch, plus or minus 0.062 inch, to provide a seat for the gasket specified in 3.11.1. The inside dimensions of the tank measured at the top shall be 15.75 inches by 4 inches, by 21.5 inches deep. A tolerance of plus or minus 0.125 inch will be permitted. A gusseted drain hole shall be provided at the bottom of each tank. The drain hole shall be provided with a solid rubber stopper, made from properly vulcanized natural rubber or synthetic rubber. Standoffs or legs shall be welded to the bottom of each tank to permit water to circulate under the tanks and to permit the tank to stand upright. The height of the standoffs shall be sufficient to provide a watertight and lighttight seal when the covers are installed on the insert tanks, and the insert tanks placed in the main tank and the main tank cover closed.

3.11.1 Insert tank covers. One insert tank cover, the same material and finish as the insert tanks, shall be provided for each insert tank. The covers shall measure 4.75 inches, plus or minus 0.062 inch, by 16.625 inches, plus or minus 0.062 inch, and shall have a 1-inch flange around the edges. The cover shall easily fit over and completely cover the open end of the insert tank. A gasket conforming to MIL-R-14328, type and class optional, shall be bonded to the cover to provide complete contact with the rim of the insert tank. One cover shall be marked "DEVELOPER" and the other cover shall be marked "FIXER". Letters shall be 0.375 inch high and shall be Arabic capital lettering. Marking shall appear in the middle of the upper surface and shall be parallel to the short side. Marking shall be stamped, etched, or engraved. Surface labeling such as printing, painting, decals, or labels attached by adhesives will not be permitted.

3.12 Film hanger. The film hanger shall be a weldment constructed of rods the same material as the main and insert tanks. The frame shall be 0.1875-inch, plus or minus 0.062 inch, rod; and the reel holders shall be 0.125-inch, plus or minus 0.062 inch, rod. The nominal outside dimensions shall be 21 inches, plus or minus 0.125 inch, in height by 14.062 inches in width. The rod shall be formed at the bottom to provide a flat base 3.5 inches, plus or minus 0.062 inch, wide to permit the hanger to stand upright. The hanger shall hold twelve film-holder reels. Film holders shall be the same material as the hanger and shall hold 70 mm film not less than 26 inches long. When filmstrips are placed in the reels, a space shall remain between each layer of film to enable processing solutions to contact all surfaces of the film reels.

3.13 Filter. A water filter with a removable cartridge shall be placed in the line between the circulating pump and the water inlet into the tank. The filter shall have sufficient capacity for the pump specified in 3.17.

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3.14 Heater. The main tank shall be equipped with a 1000-watt electric heating unit to warm the water in the main tank. The heating element shall be positioned at or near the bottom of the tank to provide uniform heating of the water.

3.15 Refrigeration system. The refrigeration system shall be a hermetically-sealed unit, and shall be mounted inside the base of the tank. The entire unit shall be removable and replaceable without loss of refrigerant and without removal of other components. The refrigerant shall conform to BB-F-1421, Type 12. The refrigeration system shall have a capacity of not less than 5650 Btu. The heater in the main tank shall be cooled by passing through a heat exchanger. A sight glass, viewable from the front when the refrigeration system is installed, shall be provided for visual inspection of the refrigerant. A normally-open solenoid valve shall be provided. The valve shall open to equalize pressure on both sides of the compressor when the refrigeration system is turned off and shall close when the compressor is operating.

3.16 High pressure control. A high pressure control conforming to MIL-C-2212 shall be installed in the high pressure side of the refrigeration piping system. The control shall stop the compressor at 185 pounds per square inch (psi) and restart the compressor at 100 psi.

3.17 Circulating pump. An electric motor-driven centrifugal pump shall be provided to circulate the water in the main tank. Water shall be taken from the bottom of the main tank to the intake of the pump, through the filter, through the heat exchanger in the refrigeration system, and discharged near the top of the main tank. Inlet and discharge pipes shall be positioned to provide good circulation of the water without splashing, and shall not interfere with the insert tanks specified in 3.11. The pump shall move water at not less than 4 gallons per minute (gpm).

3.18 Controls. All controls shall be on a panel at the front of the tank. Controls shall be as follows:

(a) Thermostat. The thermostat shall control the temperature in the main tank and shall be set at 68° F. The thermostat shall maintain the temperature of the water in the main tank at 68° F ±2° F.
(b) "ON-OFF" switch. When in the "OFF" position the switch shall prevent operation of any component of the tank.
(c) "PUMP" switch. The pump switch shall control the circulating pump only when the "ON-OFF" switch is in the "ON" position.
(d) "AUTO" switch. The automatic switch shall provide automatic operation of the heating unit and refrigeration system to maintain the correct temperature of the water in the main tank.

All controls shall be properly marked to indicate their function.

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3.19 Water pressure switch. A water pressure switch shall be provided to prevent operation of any component unless water is in the main tank. The switch shall operate as follows:

(a) Close when the water in the tank reaches 15 inches, plus or minus 2 inches.
(b) Open when the water in the tank reaches 10 inches, plus or minus 2 inches.

3.20 Time delay relays. Two time delay relays shall be provided, one for the refrigeration system and the other for the heating unit. The time delay shall be sufficiently long to preclude hunting and cycling between cooling and heating when the water in the main tank reaches operating temperature.

3.21 Moisture condensation. Any low-temperature piping of the refrigeration system or any cold-water plumbing subject to condensation shall be insulated to prevent condensation.

3.22 Tilting. The tank shall operate as specified herein when the tank is tilted up to 5 degrees from vertical in any direction.

3.23 Vibration and shock. The tank, drained of all water and chemicals, shall not be damaged or fail to operate as specified herein after testing as specified in 4.5.2.5 and 4.5.2.6.

3.24 Instruction plates. The tank shall be equipped with instruction plates including warnings, cautions, and diagrams describing any special or important procedures to be followed in assembling, operating, or servicing the tank. Plates shall conform to MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material. The plates shall be attached by bonding in conspicuous protected locations.

3.25 Identification marking. The tank shall be identified in accordance with MIL-STD-130 on an identification plate conforming to MIL-P-15024/6. The plate shall be furnished and mounted by the contractor. In addition to the identification specified in MIL-STD-130, the plate shall contain the following information:

(a) Federal stock number.
(b) Serial number.
(c) Manufacturer's name or trademark.
(d) Model number.
(e) Contract or purchase order number.
(f) Electrical rating.
(g) Refrigerant used and factory test pressure.

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3.26 Treatment and painting. The portions of the tanks normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A. Unless otherwise specified (see 6.2), the color shall match Color No. 33814 of FED. STD. No. 595. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704.

3.27 Workmanship. Workmanship shall be such that the tank shall be free from defects which detract from its appearance or which would impair its serviceability. All corners and edges exposed during use or maintenance shall be ground smooth, rounded, or otherwise treated to eliminate sharp or ragged edges.

3.27.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the tank to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.187 inch unless appropriate bridging techniques are used. The welding process used in fabrication of the tank shall be at the option of the contractor.

3.27.2 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the types of welding operations to be performed and that such qualification is effective as defined by the particular code:

Structural Welding Code, AWS D1.1, Qualification, Section 5.
Welding Qualifications of the ASME, Section IX.

Contractors who make only horizontal welds need not qualify welders for "all purpose welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with the AWS and ASME codes.

3.27.3 Welding practices.

(a) Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the contractor's recommendations.

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(b) Tack welds shall be subject to the same quality requirements as final welds.
(c) Work shall be positioned for flat welding whenever practicable.
(d) Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and the tank.

3.27.4 Cleaning. The finished tank shall be descaled and cleaned in accordance with the recommended practice of ANSI/ASTM A 380.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article tank shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article tank shall be tested as specified in 4.5.2.1 through 4.5.2.6. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

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4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1, 4.5.2.2, and 4.5.2.3. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The tanks shall be examined as specified herein for the following defects:

Major

101. Dimensions not within the limits specified. This shall include the overall dimensions of the complete tank, the main water tank, and the insert tanks.
102. Material not as specified.
103. Safety not as specified.
104. Electric wiring not as specified.
105. Mounting flange and feet not as specified.
106. Main tank cover not as specified; does not remain open by its own weight when against a back wall.
107. Separate latch for holding the main tank cover open missing, or not as specified.
108. Film hanger not as specified.
109. Low temperature piping not adequately insulated to prevent condensation.
110. Instruction plates not as specified.
111. Welding not as specified.
112. Workmanship not as specified.

Minor

201. Treatment and painting not as specified.
202. Identification marking missing or not as specified.

4.5.2 Tests.

4.5.2.1 Operational. The main tank shall be dry (no water), however, the insert tanks shall be filled to operating level with water and placed in the main tank. The tank shall be connected to a power source and the "ON-OFF" switch turned "ON". The "PUMP" switch and the "AUTO" switch shall be turned "ON". If the pump, the heating unit, or the refrigeration unit

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operate, the tank shall be rejected. Leaving the "ON-OFF" switch and "AUTO" switch "ON", the main tank shall be slowly filled with water to operating level. The tank shall be allowed to operate on "AUTO" for not less than 4 hours. Any one or more of the following shall constitute failure of this test:

(a) The refrigeration system, the heater, or the pump operates when the main tank is dry.
(b) The pressure switch fails to turn the power on when water reaches 15 inches, plus or minus 2 inches.
(c) The thermostat fails to maintain a temperature of 68° F, plus or minus 2° F.

At the end of this portion of the test, the water shall be drained with the drain valve provided. The "ON-OFF" switch and the "AUTO" switch shall be "ON". Failure of the pressure switch to shut off all operation when the water in the tank reaches a level of 10 inches, plus or minus 2 inches, shall constitute failure of this test.

4.5.2.2 Tilting. This test may be conducted during the operational test or it may be conducted as a separate test. The level of the water in the insert tanks shall be carefully measured and the lids placed on the tanks. The lids to the main tank shall be closed and latched. The tanks shall be tilted 5 degrees from the vertical, forward, backward, left, and right, and shall be operated in each position for not less than 15 minutes. During this time, the tank shall be shaken to simulate traveling over rough terrain. Failure to operate in any position as specified herein shall constitute failure of this test. At the end of this test, the water level in each insert tank shall be measured. Any evidence of leakage, i.e. more or less water in either insert tank, shall also constitute failure of this test.

4.5.2.3 Operating temperature. The tank, filled to operating level with water, shall be placed in a temperature test chamber and allowed to stabilize at the temperature specified.

4.5.2.3.1 Low temperature. The chamber shall be at plus 40° F, plus or minus 2° F. The "ON-OFF" switch and the "AUTO" switch shall be turned "ON". Failure of the heating unit to warm the water to plus 68° F, plus or minus 2° F, within 2 hours and maintain that temperature for 4 hours shall constitute failure of this test.

4.5.2.3.2 High temperature. The chamber shall be at 120° F, plus or minus 2° F. The "ON-OFF" switch and the "AUTO" switch shall be turned "ON". Failure of the refrigeration system to cool the water to plus 68° F, plus or minus 2° F, within 2 hours and maintain that temperature for 4 hours shall constitute failure of this test.

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4.5.2.4 Storage temperature. Prior to both high and low storage temperature test(s), the water shall be drained from the system by the drain valve provided. The insert tanks shall be in place but shall be empty. The cover to the main tank shall be closed and latched.

4.5.2.4.1 High temperature. The tank shall be subjected to a temperature of 160° F, plus or minus 2° F, for 4 hours per day for 3 days. At the end of this time, the tank shall be allowed to return to room temperature, and shall then be filled with water. The tank shall then be connected to a power source and operated through a complete cooling and warming cycle. Failure of any component to operate as specified herein shall constitute failure of this test.

4.5.2.4.2 Low temperature. The tank shall be subjected to a temperature of minus 30° F, plus or minus 2° F, for 12 hours with no solar radiation. The tank shall then be allowed to return to room temperature. The tank shall then be filled with water, connected to a power source, and operated through a complete cooling and warming cycle. Failure of any component to operate as specified herein shall constitute failure of this test.

4.5.2.5 Vibration. The vibration test shall be conducted at room temperature on a tank which has successfully passed the operational test specified in 4.5.2.1. The vibration test shall be conducted with the insert tanks and their lids in place and the main tank cover closed and latched. Prior to this test, all water and chemicals shall be drained from the tank. The vibration test shall be conducted in accordance with MIL-STD-810, Method 514.2, Procedure VIII. After each phase of the vibration test, the main tank and insert tanks shall be filled with water, and the tank shall be inspected and tested for proper operation. Any evidence of damage or failure to function as specified herein shall constitute failure of this test.

Note: Cycling time shall be at least 1 hour in each of three mutually perpendicular axes with 12 minutes sweep time. Dwell shall be 1/6 of the cycling time at each resonance. Acceleration shall be 2.3 g to 2.5 g through a range of 7 to 500 Hz.

4.5.2.6 Shock. The shock test shall be conducted on a tank which has successfully passed the vibration test specified in 4.5.2.5. All water and chemicals shall be drained, the insert tanks and their covers in place in the main tank, and the main tank cover shall be closed and latched. The shock test shall be conducted in accordance with MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-1 (Terminal-peak Sawtooth Shock Pulse Configuration and its Tolerance Limits). Nonconformance to 3.23 shall constitute failure of this test.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

113.	Materials, methods, or containers not as specified. Each incorrect method, material, or container shall be considered a defect.
114.	Cabinet not wrapped as specified for Level B.
115.	Cabinet not secured to container as specified for Level A or Level B.
116.	Box not provided with skids for Level A.
117.	Marking illegible, incorrect, imcomplete, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B or Commercial, as specified (see 6.2).

5.1.1 Level A. The processing tank contained in its cabinet shall be preserved in accordance with MIL-P-116, Method IIa. Technical publications shall be placed in an envelope and the envelope placed in one of the tanks.

5.1.2 Level B. Exposed metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. The cabinet shall be completely wrapped with material conforming to L-P-378, Type I, Class 1, Finish 1, grade optional, 1-1/2 mil minimum thickness. The wrap shall be secured with tape conforming to PPP-T-60. Technical publications shall be preserved Method IC-1 of MIL-P-116 and placed in one of the tanks.

5.1.3 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

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5.2.1 Level A. Each processing tank contained in its cabinet shall be packed in a wood box conforming to PPP-B-621, Class 2. The cabinet shall be secured to one side of the container by bolting through the three mounting holes provided on the back of the cabinet. The cabinet shall be mounted upright in the box and the box shall be provided with skids.

5.2.2 Level B. The tank shall be packed as specified in 5.2.1 for Level A, except that the box shall be Class 1.

5.2.3 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.3 Marking. Marking for military levels of protection shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The tank is intended for use in developing X-ray film of pipe welds. The tank is a component of the Pipeline Weld Radiographic Inspection Equipment Set.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required for inspection and approval (see 3.2, 4.3, and 6.3).
(c) Color required if other than as specified in 3.26.
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be a first produced tank. The first article should consist of one or more units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each tank.

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6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of tank. A suggested paragraph is as follows:

> "Shipment of tank shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodian:
Army - ME

Review activity:
DLA - GS

Preparing activity:
Army - ME

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-52834A(ME) Tank, Master, X-Ray Film Processing for 70 MM Film

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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